

LocalStore: Multi-Persona Operational & Screen Guide

Complete technical and operational specification covering all four ecosystem personas: **Customer**, **Merchant**, **Delivery Partner**, and **Administrator** — featuring WhatsApp Direct Connect, Coupons, In-Order Live Chat, and OTP Delivery Verification.

Platform: .NET 10 Web API + React Native (Expo SDK 52) Database: SQLite with Write-Ahead Logging (WAL Mode) Live Ports: Web UI :8081 • API :5000

Quick Access & Verification Credentials

Role / Component	Endpoint / Value	Operational Role
Frontend Web Application	http://localhost:8081 (Expo Metro)	Universal cross-persona client web app running React Native.
Backend Web API Server	http://localhost:5000/api	ASP.NET Core REST API serving catalog, orders, and settlements.
Real-Time SignalR Hub	http://localhost:5000/hubs/orders	WebSocket connection for real-time quotes, live status, and in-order chat.
Customer Persona	Mobile: 9876543210 • Password: pass123	Orders groceries, manages addresses, applies coupons, chats, and rates store.
Merchant Persona	Store: Kirana Junction • Phone: 9876500000	Dispatches shelf quotes, quick counter sales, and assigns delivery drivers.
Delivery Partner Persona	Drivers: Rohan Das (9876500001) • Aman Verma (9876500002)	Fulfills deliveries, connects on WhatsApp, and verifies 4-digit customer OTP.
Administrator Persona	Key: superadmin • Password: admin123	Oversees KYC approvals, customer standing, GMV reports, and platform fees.
Seeded Promo Coupons	FIRST50 (₹50 Off) • KIRANA10 (10% Off) • FLAT20 (₹20 Off)	Auto-seeded promotional discount codes validated via /api/coupons/validate.

Database Concurrency & Live Inspection: The SQLite database (LocalStore.db) is configured in **Write-Ahead Logging (WAL)** mode with a 30-second busy timeout. You can freely open and inspect tables in **DB Browser for SQLite** while the web app, backend, and simulated mobile clients are running simultaneously without encountering *database is locked* errors.

Global System Component: The Matrix Console Switcher

At the top of the interface sits the persistent **LocalShop Matrix Console**. Tapping any role button immediately swaps the viewport perspective without requiring logout or page reloads:

```
+-----+
|                               |
|           LocalShop Matrix Console           |
| [ Customer ] [ Merchant ] [ 🚚 Delivery (Phase 4) ] [ Admin ] |
|                               |
+-----+
```

Matrix Segment	Role Code	Target Viewport	Primary Capabilities
Customer	customer	CustomerScreen / OrderTrackingScreen	Multi-store shopping, freeform list ordering, coupon bar, live tracking, and rating.
Merchant	merchant	InventoryScreen / StaffManagementScreen	Shelf quoting, driver dispatching, counter quick billing, and bank payouts.
 Delivery	delivery	DeliveryPartnerScreen	Active delivery queue, customer WhatsApp link, phone call, and OTP verification.
Admin	admin	AdminDashboardScreen	KYC approvals, customer compliance toggling, ecosystem GMV, and audit logs.

Persona 1: Customer Operational Guide

The customer experience covers neighborhood store discovery, dual ordering modes, promotional discount coupons, real-time quote reception, live in-order chat, delivery OTP security, and post-delivery merchant reviews.

1. Multiple Delivery Addresses

Tap "Change" on the delivery banner to switch between Home, Work, or Other, or save a new address line with a tag pill.

2. WhatsApp Store Inquiry

Tap "  WhatsApp" on the active store banner to deep-link directly to the shopkeeper before placing an order.

3. Promo Coupon Engine

Apply FIRST50, KIRANA10, or FLAT20 in the cart to instantly deduct savings from your order total.

Step-by-Step Customer Journey

- 1. Authentication:** Enter your 10-digit mobile number and password on the Customer login gate. Passwords are saved with SHA-256 hashing.
- 2. Select Store & Mode:** Choose a nearby store from the carousel strip (e.g. *Kirana Junction*). Toggle between:
 - **Mode A ( Write Grocery List):** Tap daily essential chips or type what you need. Tap "Send List for Price Quote".
 - **Mode B ( Browse Catalog):** Search items, add quantities, enter a promo coupon code (e.g. FIRST50), and tap "Submit Order". Product stock automatically decrements.
- 3. Real-Time Quote Alert (SignalR):** When the merchant quotes a grocery list, the price appears instantly on the  **Track** tab without page refresh. Tap "Accept Quote & Pay" to confirm via Cash or UPI.
- 4. Delivery Confirmation OTP:** Note the prominent gold badge on your active order:  Delivery Confirmation Code: [4-Digit OTP]. Share this code with the driver on arrival.
- 5. In-Order Chat & WhatsApp:** Tap "Order Chat" for a direct live thread with the store and driver, or "  WhatsApp Store" for one-tap WhatsApp communication.
- 6. Post-Delivery 5-Star Rating:** Once the driver verifies your OTP and marks the order delivered, tap "  Rate Experience & Delivery" to submit 1-5 stars and feedback comments.

Persona 2: Merchant / Shopkeeper Operational Guide

Merchants manage grocery lists, quote prices, assign delivery staff, record walk-in counter sales, chat in real time, and withdraw digital revenue.

1. Shelf Quoting Queue

Incoming customer lists appear live. Enter the total bill amount and tap "Send Bill" to broadcast via SignalR.

2. Driver Assignment Modal

On confirmed orders, tap "Assign Driver" to pick an active staff member (e.g. Rohan Das) and transition order to *Out for Delivery*.

3. Quick Counter Billing

Record walk-in sales instantly with zero inventory maintenance by entering customer details and total amount.

Key Merchant Actions

- **Customer WhatsApp Connect:** Tap "  WhatsApp Customer" to generate an immediate WhatsApp message with the order ID, bill breakdown, and delivery verification OTP.
- **In-Order Live Chat:** Use "Order Chat" on any order card to coordinate item substitutions or out-of-stock replacements.
- **Direct Delivery:** For counter pick-ups or merchant-delivered parcels, tap "Deliver Direct" to mark completed.
- **Staff & Bank Management:** Add delivery riders under  **Staff** and dispatch instant bank payouts under  **Bank**.

 Persona 3: Delivery Partner Operational Guide (New in Phase 4)

The dedicated **Delivery Partner Console** gives drivers on the road complete visibility into assigned runs, direct customer contact channels, and secure OTP verification.

```
[ 🚚 Rohan Das (Active) ] [ 🚚 Aman Verma ] [ 🚚 All Active Drivers ]
-----
Active Run: Kirana Junction • Order ORD-29248D (Prepaid UPI)
Customer: Turja Sharma (9876543210)
Destination: Flat 4B, Greenfield Apartments, Sector 12
Items: 2x Amul Taaza Milk (500ml)
[ 💬 WhatsApp Customer ] [ 📞 Call Customer ] [ 💬 Order Chat ]
=====
[ 🔑 Verify OTP & Complete Delivery ] -> [ Enter 4-Digit Customer OTP ]
```

Delivery Partner Step-by-Step Fulfillment

- 1. **Select Driver Profile:** Switch the Matrix Switcher to 🚚 **Delivery**. Tap your driver name chip (e.g. *Rohan Das*) to load your assigned parcel queue.
- 2. **Review Delivery Destination:** Inspect customer delivery address, contact number, items list, and payment mode (Cash on Delivery vs Prepaid UPI).
- 3. **Customer Arrival Notification via WhatsApp:** Tap "💬 **WhatsApp Customer**" to trigger WhatsApp with a pre-written message: *"Hello Turja Sharma! I am your delivery partner for LocalStore Order #ORD-29248D. I am en route with your items. Please keep Delivery OTP ready."*
- 4. **Direct Voice Call:** Tap "📞 **Call Customer**" for immediate voice assistance with gate entry or apartment directions.
- 5. **Secure OTP Delivery Completion:**
 - Upon arrival, ask the customer for their 4-digit delivery confirmation code.
 - Tap "🔑 **Verify OTP & Complete Delivery**" to open the numeric verification keypad.
 - Enter the code:
 - If incorrect: System warns with an error dialog and halts fulfillment.
 - If correct: Endpoint PUT /api/orders/{id}/verify-otp-deliver verifies against SQLite, transitions status to Delivered, notifies the merchant and customer via SignalR, and logs the delivery!

 Persona 4: System Administrator Operational Guide

The Administrator oversees merchant compliance, customer security, platform fee settings, and financial volume reports.

Admin Command Tab	Database / API Target	Administrative Authority
APPROVALS	/api/admin/merchants	Inspects store registration trade licenses; issues live KYC approvals or rejections.
CUSTOMERS	/api/admin/customers	Audits registered user directory; toggles user standing between Active and Flagged.
REPORTS	/api/admin/overview	Displays real-time Gross Merchandise Volume (GMV), platform fee earnings, and order volume.
LOGS	/api/admin/logs	Immutable security ledger recording merchant disbursements, logins, and status toggles.
CONFIG	/api/admin/config	Sets global platform commission fee percentage (e.g. 2.5%) and toggles Maintenance Mode.

Complete Phase 4 API & WebSocket Hub Reference

Method	Route	Target Persona	Purpose & Persistence Target
GET / POST	/api/customers/{phone}/addresses	Customer	Lists and adds multiple saved delivery addresses (CustomerAddresses).
GET	/api/coupons	Customer / Admin	Fetches active promotional discount coupons (Coupons).
POST	/api/coupons/validate	Customer	Validates code against min order amount and calculates discount savings.
POST	/api/orders/catalog	Customer	Creates order with coupon discount; auto-generates 4-digit DeliveryOtp.
PUT	/api/orders/{id}/assign-driver	Merchant	Binds driver to order; sets status to Out for Delivery.
PUT	/api/orders/{id}/verify-otp-deliver	Delivery	Validates customer 4-digit OTP; marks order status Delivered.
PUT	/api/orders/{id}/rate	Customer	Saves 1 to 5 star rating and feedback review to SQLite Orders.
GET / POST	/api/orders/{id}/messages	All Personas	Retrieves and posts in-order live chat messages (OrderMessages).
GET	/api/orders/{id}/whatsapp-link	All Personas	Generates cross-platform https://wa.me/ deep link with prefilled order text.
PUT	/api/orders/{id}/quote	Merchant	Dispatches shelf price quote; triggers real-time SignalR push alert.
PUT	/api/orders/{id}/accept-quote	Customer	Accepts price quote and selects Cash or instant UPI payment method.
POST	/api/orders/quick-bill	Merchant	Punches counter POS sale without requiring product inventory records.
PUT	/api/orders/{id}/cancel	Customer	Cancels order, triggers instant refund, and automatically restocks items.
WebSocket	/hubs/orders	All Personas	SignalR hub broadcasting price quotes, order statuses, and live chat.

4-Persona End-to-End Test Simulation Walkthrough

Execute this 2-minute workflow across all four personas to verify end-to-end integration:

Step	Persona	Action in App UI	Observed Platform Result
1	Customer	In  Shop , add items to cart, enter FIRST50, tap " Apply ", then tap " Submit Order ".	Order placed with ₹50 discount. Stock decrements. 4-digit Delivery OTP is generated!
2	Merchant	Toggle to Merchant $\rightarrow$ Orders . Tap " Assign Driver " $\rightarrow$ choose Rohan Das .	Order status updates to Out for Delivery with driver assigned.
3	Delivery	Toggle to  Delivery $\rightarrow$ select Rohan Das . Tap "  WhatsApp Customer " or " Chat ".	WhatsApp / In-Order chat launches with customer with order details and arrival alert.
4	Customer	Toggle to Customer $\rightarrow$  Track . Review the gold banner:  Delivery Code: [OTP].	Customer views their 4-digit delivery verification OTP and driver contact.
5	Delivery	Toggle to  Delivery , tap "  Verify OTP & Complete Delivery ", enter the 4-digit OTP.	OTP verified! Order marked Delivered across all screens via SignalR.
6	Customer	Toggle to Customer $\rightarrow$  Track , tap "  Rate Experience ", submit 5 stars and review.	Store review and 5-star rating saved permanently to SQLite Orders table!
7	Admin	Toggle to Admin (superadmin / admin123), check REPORTS and LOGS .	Updated ecosystem GMV, platform commission, and audit logs verified.

Phase 4 Validation Complete: All 8 automated verification tests passed. Full cross-platform compatibility verified for mobile and web viewports.